

Applied Research Overview

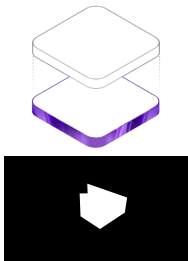
Smart i-LAB Zone 5

Robust Desk/Zone 5 occupancy estimation with imperfect live lab sensing

[pctiope/smart-i-lab-testbed](https://github.com/pctiope/smart-i-lab-testbed)

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Problem Motivation



Tracked dashboard and Desk 5 ROI assets.

Research setting

Desk/Zone 5 is a live lab zone, not a clean benchmark. Occupancy evidence arrives through camera labels, AIR-1 environmental readings, smart-plug power, mmWave motion state, SEN55 air-quality readings, and time.

- Streams can be stale, delayed, partial, or absent while the dashboard must stay useful.
- CV labels are a practical proxy for occupancy, but they inherit mask, lighting, and occlusion limits.
- A useful model must expose probability, readiness, and source health without

Research Gap

What simple approaches miss

- A dashboard can show sensor values without validating an occupancy model.
- A single-sensor rule can look stable while ignoring stale upstream data.
- A model script can optimize metrics while silently learning missingness patterns.

Gap addressed here

The Zone 5 work treats robust occupancy estimation as a multimodal, operationally gated research problem: noisy CV labels, missing sensor values, mmWave recency, rolling validation, and live deployment gates must be handled together.

Multimodal

Missingness-decoupled

Operational gates

Research Questions

1. Can a Zone 5 model combine environmental, power, mmWave, CV, and time features without treating missingness itself as occupancy evidence?
2. Can mmWave recency add short-history motion context without target leakage?
3. Can deployment and promotion gates keep the live service stable while the model contract evolves?

The deck answers these questions with source-grounded method design, architecture evidence, validation gates, and limitations rather than generated run metrics.

System And Method Overview

Method

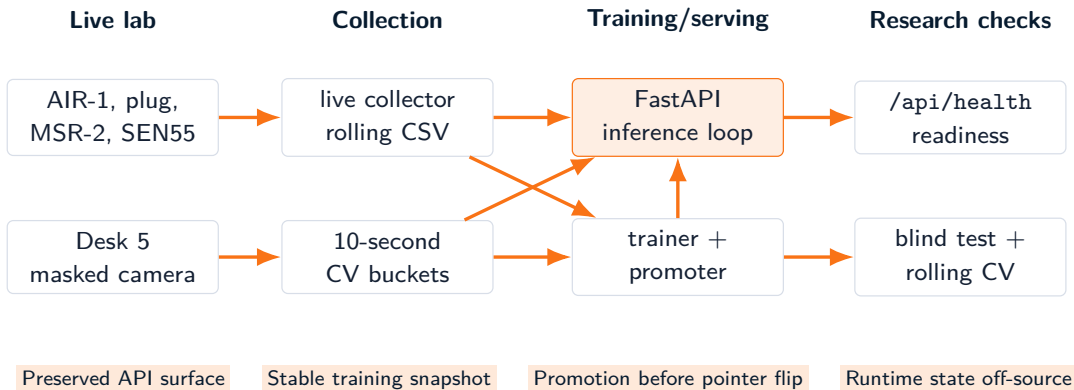
- Build a CV-derived zone_occupied target in 10-second buckets.
- Join AIR-1, power, mmWave, SEN55, and time features into rolling windows.
- Preserve raw sensor values while keeping missingness indicators out of model inputs.
- Add past-only mmWave recency features for short-history motion context.
- Deliver retrained models through staged promotion and live readiness checks.



Research principle

Model evidence and operational readiness are separated: collection can continue, the dashboard can report health, and promotion can fail without redefining the live API.

Technical Architecture As Method Evidence



Labeling And Feature Contract

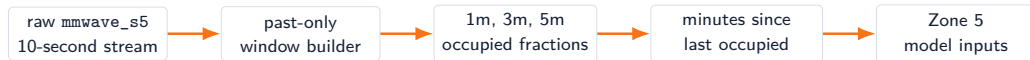
Label contract

- Target: `zone_occupied`.
- Source: Desk 5 CV count aggregated into 10-second buckets.
- Positive rule: median `occupancy_count` crosses the configured threshold.
- Missing CV buckets remain null, not false negatives.
- mmWave is an input feature, not the target.

Feature contract

- Raw inputs: AIR-1 Zone 5 environment, `power_s5`, `mmwave_s5`, `SEN55`, time.
- Model contract: `zone5_mmwave_recency_v1`.
- Missing indicators are retained as diagnostics and metadata, not model inputs.
- Core gating covers AIR-1, power, and mmWave; SEN55 is optional and neutral-filled when unavailable.

mmWave Recency Without Leakage



- Recency is derived from samples at or before the prediction window.
- No future CV label or future mmWave state is allowed into the feature window.
- The research question is whether short-history motion improves robustness.
- The guard is target separation: `zone_occupied` remains CV-derived.

Evaluation Design And Validation Gates

Offline validation

- Hidden blind-test split from the latest calendar day.
- Rolling-calendar CV on pre-test dates.
- Coverage filtering before strict fold selection.
- PR-AUC, ROC-AUC, Brier score, log loss, and both-class checks.
- Positive windows, positive buckets, and positive events as evidence gates.

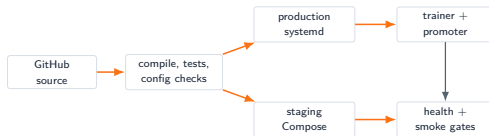
Live validation

- Smoke loading verifies target, feature columns, and artifact compatibility.
- Same-window non-regression compares candidate and production on the candidate blind-test window.
- `/api/health` exposes readiness, errors, and model pointer state.
- Promotion failure does not stop collectors or dashboard availability.

Reproducibility And Operational Validity

Why CI/CD is in the research story

Model claims are not accepted by source edits alone. Staging, production, training, and model delivery each have a separate live-exposure gate.



Boundaries

- Production: user-systemd live services.
- Staging: Docker Compose on separate ports.
- Trainer: snapshot, lock, validation, promotion.
- Delivery: vetted artifact before pointer update.

Limitations And Threats To Validity

Data and label threats

- CV labels are proxy truth and can fail under occlusion, lighting shifts, or mask errors.
- Sensor/API outages can reduce feature coverage or delay joined rows.
- Class imbalance and sparse occupied windows can make metrics unstable.
- Delayed buckets can revise recent rows after the first live append.

Operational threats

- Network reachability and upstream API latency affect live inference freshness.
- Artifact compatibility must be explicit as the contract evolves.
- A healthy dashboard is not proof that the latest candidate model is valid.
- Source docs intentionally exclude populated env values, logs, and run artifacts.

Research Roadmap And Source Anchors

Roadmap

- Continue proving strict rolling-calendar candidates before scheduled retraining is expanded.
- Track model freshness, upstream latency, and label coverage as first-class readiness signals.
- Keep raw-sensor feature evolution separate from missingness-as-signal regressions.

Topic	Primary source paths
Pipeline	zone5/.../PIPELINE.md
Features	zone5/.../feature_contract.py
CV labels	zone5/.../build_cv_training_data.py
mmWave recency	zone5/.../feature_contract.py
Validation	zone5/.../promote_model.py
Gates	.github/.../zone5-*.yaml, zone5/.../CICD_*.md

Closing point: Zone 5 treats model contract, labels, and rollout gates as part of the research method.